A Sharp Threshold Bound for Monotone Graph Properties

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Abstract

A monotone graph property is a family of graphs on a fixed vertex set that is preserved when edges are added, and that ignores the names of the vertices. As each possible edge is included independently with probability p, the probability that the random graph has the property rises from 0 to 1. The Friedgut–Kalai sharp-threshold conjecture predicts that this rise is very fast: the width of the transition is at most a constant times $(\log n)^{-2}$, where n is the number of vertices. We prove the conjecture. The proof combines two facts. First, a Fourier estimate shows that a Boolean function whose individual edges barely matter has most of its variance carried by high Fourier degrees. Second, a random "vertex block" of about $\sqrt{n}$ vertices converts information about low Fourier degrees of one restricted function into information about low Fourier degrees of the original function. Together these give a variance-versus-influence inequality, and integrating Russo's identity turns that inequality into the sharp width bound. Everything is stated and proved from scratch.

1 Introduction

Throughout, $n \geq 2$ is an integer and $[n] = \{1, 2, \ldots, n\}$. The complete graph on $[n]$ has edge set $E_n = \binom{[n]}{2}$. A graph is the same thing as a subset of E_n, or equivalently as a 0/1 vector $x = (x_e)_{e \in E_n}$ where $x_e = 1$ means edge e is present. For $0 \leq p \leq 1$, the *p-random graph* has independent edges with $\Pr(x_e = 1) = p$.

Definition 1.1. A *graph property* is a set $\mathcal{P} \subseteq \{0,1\}^{E_n}$ of graphs that is unchanged by relabelling the vertices: if $x \in \mathcal{P}$ and π is any permutation of $[n]$, then the graph with edge indicator $x_{\{\pi(u),\pi(v)\}}$ in place of $x_{\{u,v\}}$ also belongs to $\mathcal{P}$. It is *increasing* (also called *monotone*) if turning an absent edge into a present edge can never take a graph out of $\mathcal{P}$. It is *nontrivial* if it is neither empty nor the set of all graphs.

Let $\mu_p(\mathcal{P})$ be the probability that the p-random graph lies in $\mathcal{P}$. An increasing property has μ_p nondecreasing in p. For $0 < a < 1$ define the quantile

$$p_a(\mathcal{P}) = \inf\{\, p \in [0,1] : \mu_p(\mathcal{P}) \geq a \,\}.$$

All logarithms are natural, and n always counts vertices, not edges.

Theorem 1.2 (Sharp threshold)**.** *Let $n \geq 2$, let $\mathcal{P}$ be a nontrivial increasing graph property on n vertices, and let $0 < \varepsilon < 1/2$. Then*

$$p_{1-\varepsilon}(\mathcal{P}) - p_{\varepsilon}(\mathcal{P}) \leq \frac{2^{19}}{(\log n)^2} \log \frac{1}{2\varepsilon}. \tag{1.1}$$

The exponent 2 on $\log n$ is the best possible. Friedgut and Kalai observed that the property "contains a clique on $c \log n$ vertices" has a transition whose width is of order $(\log n)^{-2}$ [5, p. 2994], so no bound with a slower-decaying denominator can hold for all properties. Friedgut and Kalai proved (1.1) with denominator $\log n$ and conjectured the squared logarithm [5, Conjecture 1.2]. Bourgain and Kalai got arbitrarily close, proving a bound with denominator $(\log n)^{2-\eta}$ for every fixed $\eta > 0$ [3]. Theorem 1.2 settles the conjecture with an explicit universal constant.

The influence method goes back to Kahn, Kalai, and Linial [6], whose estimate was extended to product spaces by Bourgain, Kahn, Kalai, Katznelson, and Linial [2]; Friedgut [4] surveys this line of work. At the single unbiased measure $p = 1/2$, Kelman, Kindler, Lifshitz, Minzer, and Safra proved $I_{1/2}(f) \geq c\,(\log n)^2/(\log\log n)^2 \operatorname{Var}_{1/2}(f)$ for all large n and an absolute constant $c > 0$ [7]; that controls one bias, whereas a threshold-width bound must hold across the whole transition, which is what the variance estimate below supplies.

The engine of the proof is a variance bound. For a Boolean function f on graphs, its *pivotal influence* at edge e is the probability that toggling e alone changes f, and its *total influence* is the sum over edges. Write $x_{e\leftarrow b}$ for the graph x with edge e forced to b, and let the probability be over all edges other than e.

Definition 1.3. For $f : \{0,1\}^{E_n} \to \{0,1\}$ and $0 < p < 1$,

$$I_{e,p}(f) = \Pr_p(f(x_{e\leftarrow 1}) \neq f(x_{e\leftarrow 0})), \qquad I_p(f) = \sum_{e \in E_n} I_{e,p}(f).$$

All probabilities and expectations in the subscript p use the independent-edge measure with $\Pr(x_e = 1) = p$.

Theorem 1.4 (Variance controls influence)**.** *Let $n \geq 2$, let $f : \{0,1\}^{E_n} \to \{0,1\}$ be invariant under vertex permutations, and let $0 < p < 1$. Then*

$$\operatorname{Var}_p(f) \leq \frac{2^{17}}{(\log n)^2}\, I_p(f). \tag{1.2}$$

No monotonicity is assumed.

The proof of Theorem 1.2 from Theorem 1.4 is short and is done in Section 4. The rest of the paper proves Theorem 1.4.

How the variance bound is proved. Think of f as a function on the edges of the complete graph, and expand it in the Fourier basis adapted to the biased measure. Two different notions of "degree" then appear. There is the *Fourier degree* $|S|$, the number of edges in a Fourier term, and there is the *conditional Fourier degree* of f after some edges are frozen. The difficult part of any sharp-threshold argument is that these two notions behave very differently. A vertex-symmetric function may have huge total influence even when no single edge matters much, and then it is the high Fourier degrees that soak up the variance.

The argument has two independent pieces.

1. *A Fourier lemma* (Section 2). If every single-edge influence of a Boolean h is at most $1/m$ of its total influence, then the Fourier coefficients of degrees between 1 and about $\sigma \log m$ carry at most $m^{-1/4}$ times the total influence. Here $\sigma = \sqrt{p(1-p)}$ measures the bias. The point is that the bound is *linear* in the total influence, so it can later be averaged.

2. *A geometric restriction* (Section 3). Pick a random set B of about $\sqrt{n}$ vertices and freeze every edge that meets B in neither endpoint. Permutations of B survive the freezing, so piece 1 applies to each restricted function. The gain is a dictionary between degrees: a Fourier term S of the original function survives with conditional degree equal to $|S \cap F_B|$, where F_B is the set of edges touching B. Because any small edge set has a low-degree vertex, we can choose B so that only that vertex of S falls in B, and the surviving degree is small exactly when $|S|$ is not too large.

Combining the two gives (1.2); Section 4 finishes the argument.

We record for the reader the two companion papers that treat different threshold questions and are not used here: the equivalence of integral and fractional expectation thresholds [8], and the second Kahn–Kalai conjecture [9].

2 A Fourier estimate for Boolean functions with flat influence

This section proves one analytic lemma. It says that if a Boolean function depends on many coordinates in a balanced way, so that no single coordinate carries more than a $1/m$ fraction of the total influence, then its low-degree Fourier coefficients carry very little of the L^2 mass. The statement is linear in the total influence, which is exactly what the averaging in Section 3 will need.

The biased Fourier basis. Fix a finite set J of coordinates and a bias $0 < p < 1$, and put

$$\sigma = \sqrt{p(1-p)}, \qquad 0 < \sigma \le \tfrac{1}{2}.$$

All expectations and norms in this section are with respect to the product measure on $\{0,1\}^J$ in which each coordinate equals 1 with probability p. For $q \ge 1$ write $\|g\|_q = (\mathbb{E}|g|^q)^{1/q}$. Define, for $i \in J$,

$$\chi_i(x) = \frac{x_i - p}{\sigma}, \qquad \chi_S(x) = \prod_{i \in S} \chi_i(x) \quad (S \subseteq J), \qquad \chi_\varnothing = 1.$$

Each χ_i has mean 0 and second moment 1. Since the coordinates are independent, the $2^{|J|}$ functions χ_S are orthonormal:

$$\mathbb{E}[\chi_S \chi_T] = \mathbf{1}\{S = T\}.$$

They are $2^{|J|}$ orthonormal vectors in the $2^{|J|}$-dimensional space of real functions on $\{0,1\}^J$, so they form an orthonormal basis. Writing the Fourier coefficient of g at S as $\widehat{g}(S) = \mathbb{E}[g\,\chi_S]$, we get the expansion and Parseval identity

$$g = \sum_{S \subseteq J} \widehat{g}(S)\,\chi_S, \qquad \mathbb{E}g^2 = \sum_{S \subseteq J} \widehat{g}(S)^2, \qquad \mathrm{Var}(g) = \sum_{\varnothing \ne S \subseteq J} \widehat{g}(S)^2. \tag{2.1}$$

For $i \in J$, let $\Delta_i g(x) = g(x_{J\setminus\{i\}}, 1) - g(x_{J\setminus\{i\}}, 0)$, a function of the coordinates other than i. Averaging the product $g\,\chi_i$ over the single coordinate i leaves a factor $p(1-p)/\sigma = \sigma$ times this difference:

$$\mathbb{E}_i[g\,\chi_i] = \frac{p(1-p)}{\sigma}\,\Delta_i g = \sigma\,\Delta_i g.$$

Consequently, for $A \subseteq J \setminus \{i\}$, the coefficient of g at $A \cup \{i\}$ equals σ times the coefficient of $\Delta_i g$ at A (the latter taken on the coordinates $J \setminus \{i\}$):

$$\widehat{g}(A \cup \{i\}) = \sigma\,\widehat{\Delta_i g}(A). \tag{2.2}$$

For a Boolean $h : \{0,1\}^J \to \{0,1\}$, define the influence of coordinate i and the total influence by

$$I_i(h) = \Pr(\Delta_i h \neq 0), \qquad I(h) = \sum_{i \in J} I_i(h).$$

Because $\Delta_i h \in \{-1, 0, 1\}$, it satisfies $|\Delta_i h| = \mathbf{1}\{\Delta_i h \neq 0\}$ and so $\|\Delta_i h\|_{4/3} = I_i(h)^{3/4}$. Applying Parseval on the coordinates $J \setminus \{i\}$ to (2.2) gives the derivatives of the squared mass, and summing over i gives the two identities

$$\sum_{S \ni i} \widehat{h}(S)^2 = \sigma^2 I_i(h), \qquad \sum_{S \subseteq J} |S|\, \widehat{h}(S)^2 = \sigma^2 I(h). \tag{2.3}$$

These hold for every Boolean h; no monotonicity is used anywhere in this section.

The noise operator and hypercontractivity. The proof of the low-degree bound goes through a smoothing operator. For $0 < \rho \leq 1$ define the linear operator T_ρ on functions on $\{0,1\}^J$ by its action on the basis,

$$T_\rho \chi_S = \rho^{|S|} \chi_S.$$

In other words, T_ρ multiplies each Fourier coefficient of degree d by ρ^d. The following lemma is the standard hypercontractive estimate of Bonami [1], proved here for a general bias so that the dependence on σ is explicit.

Lemma 2.1 (Hypercontractive norm bound)**.** *For every real function g on $\{0,1\}^J$ and $\rho = \sigma/4$,*

$$\|T_\rho g\|_4 \leq \|g\|_2.$$

Proof. Since $0 < \sigma \leq 1/2$ we have $0 < \rho \leq 1/8$. It is enough to handle a single coordinate and then tensorize.

One coordinate. The bound $|\chi_i| \leq \sigma^{-1}$ and $\mathbb{E}\chi_i^2 = 1$ give $|\mathbb{E}\chi_i^3| \leq \sigma^{-1}$ and $\mathbb{E}\chi_i^4 \leq \sigma^{-2}$. For real numbers a, b, expand $\mathbb{E}(a + \rho b\, \chi_i)^4$ and bound the odd term using $2|ab^3| \leq a^2 b^2 + b^4$:

$$\begin{aligned}
\mathbb{E}(a + \rho b\, \chi_i)^4 &= a^4 + 6\rho^2 a^2 b^2 + 4\rho^3 a b^3\, \mathbb{E}\chi_i^3 + \rho^4 b^4\, \mathbb{E}\chi_i^4 \\
&\leq a^4 + 6\rho^2 a^2 b^2 + \frac{4\rho^3}{\sigma} |ab^3| + \frac{\rho^4}{\sigma^2} b^4 \\
&\leq a^4 + 6\rho^2 a^2 b^2 + \frac{2\rho^3}{\sigma}(a^2 b^2 + b^4) + \frac{\rho^4}{\sigma^2} b^4.
\end{aligned}$$

Substituting $\rho = \sigma/4$, the coefficient of $a^2 b^2$ is $6\rho^2 + 2\rho^3/\sigma = 6\sigma^2/16 + 2\sigma^2/64 = 13\sigma^2/32$, and the coefficient of b^4 is $2\rho^3/\sigma + \rho^4/\sigma^2 = \sigma^2/32 + \sigma^2/256 = 9\sigma^2/256$. Hence

$$\mathbb{E}(a + \rho b\, \chi_i)^4 \leq a^4 + \frac{13\sigma^2}{32} a^2 b^2 + \frac{9\sigma^2}{256} b^4 \leq (a^2 + b^2)^2,$$

where the last step uses $\sigma^2 \leq 1/4$, which makes $\frac{13}{32}\sigma^2 \leq \frac{13}{128} < 2$ and $\frac{9}{256}\sigma^2 \leq \frac{9}{1024} < 1$.

Tensorization. Induct on $|J|$; the empty product is trivial. Pick a coordinate i and write $g = a + b\, \chi_i$ where a, b depend only on the other coordinates. Let $A = T_\rho a$ and $B = T_\rho b$ be the images under T_ρ on those other coordinates, so that $T_\rho g = A + \rho B\, \chi_i$. Fix the other coordinates; the single-coordinate inequality, applied to the numbers A and B, gives

$$\mathbb{E}_i\big[(T_\rho g)^4\big] = \mathbb{E}_i(A + \rho B\, \chi_i)^4 \leq (A^2 + B^2)^2.$$

Let $X = \mathbb{E}_i[(T_\rho g)^4]$, a function of the other coordinates, so $X \le (A^2 + B^2)^2$ pointwise. Since $\|T_\rho g\|_4^2 = (\mathbb{E}X)^{1/2}$ and $\|f^2\|_1 = \|f\|_2^2$,

$$\|T_\rho g\|_4^2 \le \|A^2 + B^2\|_2 \le \|A\|_4^2 + \|B\|_4^2 \le \|a\|_2^2 + \|b\|_2^2 = \|g\|_2^2,$$

where the third step is the induction hypothesis applied to a and b, and the last equality is Parseval, valid because a, b are functions of the coordinates other than χ_i. Taking square roots gives $\|T_\rho g\|_4 \le \|g\|_2$. □

From the norm bound to low degrees. Let $P_{\le d}$ denote the orthogonal projection onto the span of the characters of degree at most d, i.e. $P_{\le d} g = \sum_{|S| \le d} \widehat{g}(S)\chi_S$. On that span, T_ρ is invertible with inverse given by $T_\rho^{-1}\chi_S = \rho^{-|S|}\chi_S$. If $u = P_{\le d}u$, then by Parseval

$$\|T_\rho^{-1}u\|_2^2 = \sum_{|S| \le d} \rho^{-2|S|}\widehat{u}(S)^2 \le \rho^{-2d}\|u\|_2^2. \tag{2.4}$$

Applying Lemma 2.1 to $T_\rho^{-1}u$ gives $\|u\|_4 = \|T_\rho(T_\rho^{-1}u)\|_4 \le \|T_\rho^{-1}u\|_2 \le \rho^{-d}\|u\|_2$. Finally, duality in the finite-dimensional subspace and Holder's inequality give, for every real function v,

$$\|P_{\le d}v\|_2 = \sup_{\substack{u = P_{\le d}u \\ \|u\|_2 = 1}} \big|\mathbb{E}[vu]\big| \le \rho^{-d}\|v\|_{4/3}. \tag{2.5}$$

Lemma 2.2 (Low-degree bound)**.** *Let $m \ge 2$ be real and let $h : \{0,1\}^J \to \{0,1\}$ satisfy $I_i(h) \le I(h)/m$ for every $i \in J$. Put*

$$k = \frac{\sigma \log m}{64}.$$

Then

$$\sum_{1 \le |S| \le k} \widehat{h}(S)^2 \le m^{-1/4}\, I(h). \tag{2.6}$$

Degrees are integers $|S|$, so the sum is empty (and the bound trivial) when $k < 1$.

Proof. If $k < 1$ the sum is empty. Assume $k \ge 1$. Every nonempty Fourier set is counted at least once when we sum over its coordinates, so

$$\sum_{1 \le |S| \le k} \widehat{h}(S)^2 \le \sum_{i \in J} \sum_{\substack{S \ni i \\ |S| \le k}} \widehat{h}(S)^2.$$

Fix i. By (2.2), $\widehat{h}(S)^2$ for $S = A \cup \{i\}$ with $|A| = |S| - 1 \le k - 1$ equals $\sigma^2 \widehat{\Delta_i h}(A)^2$. The inner sum is therefore σ^2 times the squared L^2 norm of the projection of $\Delta_i h$ onto degrees at most $k-1$ on $J \setminus \{i\}$. Applying (2.5) with $d = k - 1 \le k$ and $v = \Delta_i h$,

$$\sum_{\substack{A \subseteq J \setminus \{i\} \\ |A| \le k-1}} \widehat{\Delta_i h}(A)^2 \le \rho^{-2(k-1)}\|\Delta_i h\|_{4/3}^2 \le \rho^{-2k}\, I_i(h)^{3/2}.$$

Summing over i and using (2.2)'s factor σ^2,

$$\sum_{1 \le |S| \le k} \widehat{h}(S)^2 \le \sigma^2 \rho^{-2k} \sum_{i \in J} I_i(h)^{3/2}. \tag{2.7}$$

We bound the two factors. First, since $\rho = \sigma/4$,

$$\rho^{-2k} = \Big(\frac{4}{\sigma}\Big)^{2k} = \exp\Big(2k \log \frac{4}{\sigma}\Big) = \exp\Big(\frac{\sigma \log m}{32} \log \frac{4}{\sigma}\Big) = m^{\sigma \log(4/\sigma)/32}.$$

The function $s \mapsto s \log(4/s)$ is increasing on $(0, 1/2]$, because its derivative $\log(4/s) - 1$ is positive there. Hence, as $\sigma \le 1/2$,

$$\sigma \log \frac{4}{\sigma} \le \tfrac{1}{2} \log 8 < 2, \qquad \text{so} \qquad \rho^{-2k} \le m^{1/16}.$$

Second, split on the size of $I(h)$. If $I(h) \le m^{1/4}$, then $I_i(h) \le I(h)/m$ gives $I_i(h)^{3/2} = I_i(h)\sqrt{I_i(h)} \le I_i(h)\sqrt{I(h)/m}$, and summing over i,

$$\sum_{i \in J} I_i(h)^{3/2} \le \sqrt{\frac{I(h)}{m}}\, I(h) \le m^{-3/8} I(h).$$

Inserting both bounds into (2.7) and using $\sigma^2 \le 1/4 \le 1$,

$$\sum_{1 \le |S| \le k} \widehat{h}(S)^2 \le \sigma^2\, m^{1/16}\, m^{-3/8} I(h) = \sigma^2\, m^{-5/16} I(h) \le m^{-1/4} I(h),$$

since $-5/16 < -1/4$. If instead $I(h) > m^{1/4}$, then by Parseval and the Boolean range $0 \le h \le 1$,

$$\sum_{1 \le |S| \le k} \widehat{h}(S)^2 \le \mathbb{E} h^2 \le 1 < m^{-1/4}\, I(h).$$

Either way (2.6) holds, for every $0 < p < 1$ and with no lower bound on σ. □

3 Restricting to the edges incident with a vertex block

We now exploit the vertex symmetry of f. Fix $p \in (0, 1)$ and $\sigma = \sqrt{p(1-p)}$; from here to the end of the section all Fourier coefficients, variances, and influences use the bias p on $E_n = \binom{[n]}{2}$, and $f : \{0,1\}^{E_n} \to \{0,1\}$ is invariant under every permutation of $[n]$. Monotonicity is never used.

Two kinds of degree. Lemma 2.2 controls Fourier degrees of a restricted function, but we need a statement about Fourier degrees of f itself. The bridge is a random set B of vertices. Consider the set of edges

$$F_B = \{e \in E_n : e \cap B \neq \varnothing\}$$

that touch B; these are the *free* edges. We freeze every edge in the complement $H_B = E_n \setminus F_B$, i.e. every edge with both endpoints outside B. For a fixed freezing $y \in \{0,1\}^{H_B}$, let f_y be the resulting function of the free coordinates $x \in \{0,1\}^{F_B}$, namely $f_y(x) = f(x, y)$.

Two things make this choice useful. First, a permutation of the vertices of B, extended by the identity elsewhere, fixes every edge of H_B (they avoid B), so it fixes y and acts only on free coordinates. Thus f_y inherits the full vertex symmetry of f on the block B, even when the frozen graph y has no symmetry at all. Second, the free set is not a thin slice of the edges: for a Fourier support S for which B meets the nonisolated vertex set $V(S)$ in exactly one vertex u, the surviving part $S \cap F_B$ consists exactly of the edges of S incident with u, so its size is the graph degree $d_S(u)$.

The plan is therefore: pick B so that it captures a single low-degree vertex of S. If S has s edges, some nonisolated vertex of S has degree at most $\sqrt{2s}$, so a set S with $s \le k^2/2$ has a vertex of degree at most k. Then the conditional Fourier degree of f_y at S is at most k, and Lemma 2.2 applies. We make this precise below.

Exact averaging over a restriction. The first tool is an identity relating the Fourier coefficients of the restricted functions f_y to those of f. It holds for every Boolean f and uses no symmetry.

Lemma 3.1 (Exact averaging over a restriction)**.** *Let $F \subseteq E_n$, put $H = E_n \setminus F$, and for $y \in \{0,1\}^H$ let $f_y(x) = f(x,y)$ with $x \in \{0,1\}^F$. If y has the Bernoulli-p product law, then for every real $k > 0$,*

$$\mathbb{E}_y \sum_{\substack{A \subseteq F \\ 1 \le |A| \le k}} \widehat{f_y}(A)^2 = \sum_{\substack{S \subseteq E_n \\ 1 \le |S \cap F| \le k}} \widehat{f}(S)^2, \tag{3.1}$$

$$\mathbb{E}_y\, I(f_y) = \sum_{e \in F} I_e(f). \tag{3.2}$$

Proof. Expand f in the product Fourier basis on E_n and integrate the free coordinates x. For each $A \subseteq F$ this gives the exact finite identity

$$\widehat{f_y}(A) = \sum_{D \subseteq H} \widehat{f}(A \cup D)\, \chi_D(y),$$

because the product $\chi_{A \cup D}$ splits as $\chi_A(x)\chi_D(y)$ and χ_A is averaged out against the coefficient. Orthonormality of the χ_D on the exterior coordinates H then gives $\mathbb{E}_y \widehat{f_y}(A)^2 = \sum_{D \subseteq H} \widehat{f}(A \cup D)^2$. Summing over all A with $1 \le |A| \le k$ produces (3.1). Notice that the exterior degree $|D|$ is completely unconstrained: $S = A \cup D$ may have any size as long as $|S \cap F| = |A|$ lies between 1 and k.

For the influence identity, note that for $e \in F$ the pivotal event of f_y at e is the pivotal event of f at e with the exterior coordinates fixed to y. Averaging over y and over the remaining free coordinates counts the same pivotal event of f, so $\mathbb{E}_y I_e(f_y) = I_e(f)$ for each $e \in F$. Summation over $e \in F$ proves (3.2). □

The block estimate. Fix m with $3 \le m \le n$, and let B be uniformly random among the m-element subsets of $[n]$. The free set is F_B, as above. The point of the next proposition is to apply Lemma 2.2 to f_y for each freezing y and then average.

Proposition 3.2 (The block estimate)**.** *Let $3 \le m \le n$ be an integer, let B be uniform among the m-element subsets of $[n]$, and set $F_B = \{e : e \cap B \neq \varnothing\}$ together with*

$$k = \frac{\sigma \log m}{64}.$$

Then

$$\sum_{S \subseteq E_n} \Pr_B\{1 \le |S \cap F_B| \le k\}\, \widehat{f}(S)^2 \le \frac{2m}{n}\, m^{-1/4}\, I(f). \tag{3.3}$$

Proof. Fix B, and fix an arbitrary assignment y to $H_B = E_n \setminus F_B$. The free set F_B contains *all* edges with at least one endpoint in B: the $\binom{m}{2}$ edges inside B and, for each vertex $w \notin B$, the m edges from w to B. Every permutation of B, extended by the identity on $[n] \setminus B$, fixes each edge of H_B individually and hence fixes y. So f_y is invariant under the permutation group of B, even if the graph encoded by y has no symmetry.

These permutations partition F_B into orbits: one orbit of size $\binom{m}{2}$ (the edges inside B) and, for each fixed $w \in [n] \setminus B$, one orbit of size m (the edges $\{u, w\}$, $u \in B$). Since $m \ge 3$, every orbit has size at least m. The measure is a product law and pivotal events are permuted, so $I_e(f_y)$

is constant on each orbit. If the orbits are $O_1, \dots, O_t$ with sizes $s_j \geq m$ and common influence $c_j = I_e(f_y)$ for $e \in O_j$, then

$$I(f_y) = \sum_{j=1}^{t} s_j c_j \geq s_j c_j \geq m\, c_j,$$

so $c_j \leq I(f_y)/m$. Hence

$$I_e(f_y) \leq \frac{I(f_y)}{m} \qquad (e \in F_B).$$

Now apply Lemma 2.2 to the restricted function f_y on the coordinates F_B: its hypothesis holds, and its conclusion reads

$$\sum_{\substack{A \subseteq F_B \\ 1 \leq |A| \leq k}} \widehat{f_y}(A)^2 \leq m^{-1/4}\, I(f_y).$$

The right-hand side is linear in $I(f_y)$, so averaging over y and using Lemma 3.1 gives

$$\sum_{\substack{S \subseteq E_n \\ 1 \leq |S \cap F_B| \leq k}} \widehat{f}(S)^2 \leq m^{-1/4} \sum_{e \in F_B} I_e(f).$$

Finally average over B. A fixed edge e lies in F_B exactly when at least one of its two endpoints lies in B, which happens with probability at most $2m/n$ by a union bound. Thus $\mathbb{E}_B \sum_{e \in F_B} I_e(f) = \sum_e \Pr_B(e \in F_B) I_e(f) \leq \frac{2m}{n} I(f)$, and (3.3) follows. □

Capturing low Fourier degrees. Proposition 3.2 only sees Fourier terms whose intersection with F_B is a between 1 and k edges. The next lemma shows that this happens with uniformly positive probability for every small support S, by choosing B to meet $V(S)$ at a single low-degree vertex.

Lemma 3.3 (Capturing low Fourier levels)**.** *Suppose $t = \log n \geq 16$, and put*

$$m = \lfloor \sqrt{n} \rfloor, \qquad k = \frac{\sigma \log m}{64}.$$

Then

$$\sum_{\substack{S \subseteq E_n \\ 1 \leq |S| \leq k^2/2}} \widehat{f}(S)^2 \leq 4\, m^{-1/4}\, I(f). \tag{3.4}$$

Proof. Fix a nonempty $S \subseteq E_n$ with $s = |S| \leq k^2/2$, and view S as the edge set of a simple graph. Let $V(S)$ be its set of nonisolated vertices and write $v = |V(S)|$. Since $s \leq \binom{v}{2} < v^2/2$, the average of d_S over $V(S)$ equals $2s/v < \sqrt{2s}$, so some nonisolated vertex $u \in V(S)$ has degree at most this average:

$$1 \leq d_S(u) \leq \sqrt{2s} \leq k, \qquad v \leq 2s \leq k^2. \tag{3.5}$$

Choose B uniformly among the m-element vertex sets. The event $B \cap V(S) = \{u\}$ has $u \in B$ and no other vertex of $V(S)$ in B. Given $u \in B$, the other $m-1$ members of B are a random $(m-1)$-subset of the $n-1$ remaining vertices, so a union bound gives

$$\Pr_B\{B \cap V(S) = \{u\}\} \geq \frac{m}{n}\Big(1 - (v-1)\frac{m-1}{n-1}\Big). \tag{3.6}$$

On this event the set $S \cap F_B$ consists exactly of the edges of S incident with u, so $|S \cap F_B| = d_S(u)$, which lies in $[1, k]$ by (3.5).

We bound the probability uniformly in p. First $k \le t$, because $k = \sigma \log m/64 \le \frac{1}{2} \log m/64 \le \frac{1}{2}(t/2)/64 < t$. Hence $v \le k^2 \le t^2$. Also

$$\frac{m-1}{n-1} \le \frac{1}{\sqrt{n}},$$

since $m \le \sqrt{n}$ gives $m - 1 \le \sqrt{n} - 1 = \frac{n-1}{\sqrt{n}+1} \le \frac{n-1}{\sqrt{n}}$. Therefore

$$(v-1)\frac{m-1}{n-1} \le t^2 \cdot \frac{1}{\sqrt{n}} = t^2 e^{-t/2} \le 256\, e^{-8} < \tfrac{1}{2},$$

where $t^2 e^{-t/2}$ is decreasing for $t \ge 16$ and so is at most its value $256e^{-8}$ at $t = 16$. Combining (3.6) with this bound,

$$\Pr_B\{1 \le |S \cap F_B| \le k\} \ge \frac{m}{n} \cdot \frac{1}{2} = \frac{m}{2n}, \tag{3.7}$$

for every support under consideration.

Now apply (3.7) summand by summand. The choice of u depends on S, but the bound is uniform, so for each $\widehat{f}(S)^2 \ge 0$,

$$\widehat{f}(S)^2 \le \frac{2n}{m} \Pr_B\{1 \le |S \cap F_B| \le k\} \widehat{f}(S)^2.$$

Summing over the finitely many S with $1 \le |S| \le k^2/2$, comparing with (3.3), and cancelling the common factor,

$$\sum_{\substack{S \subseteq E_n \\ 1 \le |S| \le k^2/2}} \widehat{f}(S)^2 \le \frac{2n}{m} \cdot \frac{2m}{n}\, m^{-1/4} I(f) = 4\, m^{-1/4} I(f).$$

If $k^2/2 < 1$ the sum is empty and the conclusion is vacuous. □

Proof of the variance bound. We now assemble the two estimates.

Proof of Theorem 1.4. First suppose $t = \log n \ge 16$, and take $m = \lfloor \sqrt{n} \rfloor$ and $k = \sigma \log m/64$ as in Lemma 3.3; then $m \ge 3$ and $k > 0$. The high Fourier degrees are controlled directly from the Dirichlet identity (2.3). Since $|S| > k^2/2$ implies $1 < 2|S|/k^2$,

$$\sum_{|S| > k^2/2} \widehat{f}(S)^2 \le \frac{2}{k^2} \sum_{S \subseteq E_n} |S|\, \widehat{f}(S)^2 = \frac{2\sigma^2}{k^2}\, I(f). \tag{3.8}$$

Combine Parseval (2.1), the captured mass (3.4), and this tail (3.8):

$$\mathrm{Var}(f) = \sum_{\varnothing \ne S} \widehat{f}(S)^2 \le \Big(4\, m^{-1/4} + \frac{2\sigma^2}{k^2}\Big) I(f). \tag{3.9}$$

It remains to bound the bracketed constant. Because $m \ge \sqrt{n}/2$ and $t \ge 16$, we have $\log m \ge t/2 - \log 2 \ge t/3$, so

$$\frac{2\sigma^2}{k^2} = \frac{2\sigma^2}{(\sigma \log m/64)^2} = \frac{8192}{(\log m)^2} \le \frac{73728}{t^2},$$

an exact cancellation of σ^2 that holds even when p is close to 0 or 1. For the other term, $m^{-1/4} \le (\sqrt{n}/2)^{-1/4} = 2^{1/4} n^{-1/8} = 2^{1/4} e^{-t/8}$, and $t^2 e^{-t/8}$ is nonincreasing for $t \ge 16$, so

$$4\, m^{-1/4} t^2 \le 4 \cdot 2^{1/4}\, t^2 e^{-t/8} \le 4 \cdot 2^{1/4} \cdot 256\, e^{-2} < 256.$$

Thus $4m^{-1/4} \le 256/t^2$ and the right-hand side of (3.9) is at most $(73728 + 256) I(f)/t^2 \le 2^{17} I(f)/(\log n)^2$, proving (1.2).

If instead $2 \le n < e^{16}$, the Fourier identities give immediately

$$\mathrm{Var}(f) \le \sigma^2 I(f) \le \tfrac{1}{4} I(f) \le \frac{64}{(\log n)^2} I(f),$$

because $\sigma^2 \le 1/4$ and $(\log n)^2 < 256$. Hence the single constant $C_0 = 2^{17}$ works for every integer $n \ge 2$ and every $p \in (0,1)$. □

4 From influence to threshold width

This short section turns Theorem 1.4 into Theorem 1.2 by integrating the probability derivative. The key formula is due to Russo [10, Lemma 3] and is used in the same way by Friedgut and Kalai [5, Lemma 2.2].

Proof of Theorem 1.2. Let $f = \mathbf{1}_{\mathcal{P}}$ be the indicator of an increasing graph property, let $\mu(p) = \mathbb{E}_p f$ be the probability that the p-random graph lies in $\mathcal{P}$, and put $C_0 = 2^{17}$.

Russo's identity. If the edge probabilities are allowed to vary independently, $\mathbb{E} f$ is a multilinear polynomial in the $|E_n|$ individual edge probabilities. Differentiating in the probability of a single edge e gives the expectation of the discrete difference $\Delta_e f = f(x_{e\leftarrow 1}) - f(x_{e\leftarrow 0})$, all other coordinates retaining their probabilities. For an increasing f, the difference $f(x_{e\leftarrow 1}) - f(x_{e\leftarrow 0})$ takes only the values 0 and 1, so its expectation, with all other biases set to p, is exactly the pivotal influence $I_{e,p}(f)$ of Definition 1.3. Summing the partial derivatives along the diagonal where every edge has probability p therefore gives

$$\mu'(p) = I_p(f). \tag{4.1}$$

The differential inequality. Since f is Boolean, $\mathrm{Var}_p(f) = \mu(p)(1-\mu(p))$. Theorem 1.4 together with (4.1) gives

$$\mu'(p) \;\ge\; \frac{(\log n)^2}{C_0}\, \mu(p)(1-\mu(p)) \qquad (0 < p < 1). \tag{4.2}$$

Quantiles are well defined. Nontriviality and monotonicity give $f(\varnothing) = 0$ and $f(E_n) = 1$, and both graphs have positive probability under every interior bias, so $0 < \mu(p) < 1$ for $0 < p < 1$ while $\mu(0) = 0$ and $\mu(1) = 1$. By continuity μ has a unique interior crossing $\mu(p_a) = a$ for every $0 < a < 1$. Thus each quantile p_a is a genuine interior point, and on the interval $[p_\varepsilon, p_{1-\varepsilon}]$ the division in (4.2) is legitimate:

$$\frac{d}{dp} \log \frac{\mu(p)}{1-\mu(p)} = \frac{\mu'(p)}{\mu(p)(1-\mu(p))} \ge \frac{(\log n)^2}{C_0}.$$

Integration. The log-odds rise with slope at least $(\log n)^2/C_0$ across $[p_\varepsilon, p_{1-\varepsilon}]$. At the endpoints the odds are $(1-\varepsilon)/\varepsilon$ and $\varepsilon/(1-\varepsilon)$, so

$$2 \log \frac{1-\varepsilon}{\varepsilon} = \log \frac{1-\varepsilon}{\varepsilon} - \log \frac{\varepsilon}{1-\varepsilon} \ge \frac{(\log n)^2}{C_0} (p_{1-\varepsilon} - p_\varepsilon),$$

that is,

$$p_{1-\varepsilon} - p_\varepsilon \le \frac{2C_0}{(\log n)^2} \log \frac{1-\varepsilon}{\varepsilon}.$$

Finally $4\varepsilon(1-\varepsilon) \le 1$ gives $\log \frac{1-\varepsilon}{\varepsilon} \le 2 \log \frac{1}{2\varepsilon}$, hence

$$p_{1-\varepsilon} - p_\varepsilon \le \frac{4C_0}{(\log n)^2} \log \frac{1}{2\varepsilon} = \frac{2^{19}}{(\log n)^2} \log \frac{1}{2\varepsilon},$$

which is (1.1). □

The equivalent additive form. Friedgut and Kalai state the conjecture as an additive statement about a single probability. Setting

$$\delta = \frac{2^{19}}{(\log n)^2} \log \frac{1}{2\varepsilon}$$

recovers it. If $\mu(p) > \varepsilon$ then $p > p_\varepsilon$, so Theorem 1.2 gives $p+\delta > p_{1-\varepsilon}$ and therefore $\mu(p+\delta) > 1-\varepsilon$ whenever $p+\delta \le 1$; when $p+\delta > 1$ the endpoint already has probability 1. Conversely, if $p_\varepsilon+\delta < 1$, apply this to any $p > p_\varepsilon$ and let $p \downarrow p_\varepsilon$; continuity gives $\mu(p_\varepsilon + \delta) \ge 1 - \varepsilon$, which is the same bound. When $p_\varepsilon + \delta \ge 1$ the bound is automatic.

The transition near its median. The log-odds estimate also describes the profiles of μ on both sides of the median $p_* = p_{1/2}$, where the log-odds vanish. Integrating (4.2) outward from p_* gives, with $C_0 = 2^{17}$,

$$\mu(p_* - s) \le \frac{1}{1 + \exp\bigl(s(\log n)^2/C_0\bigr)} \quad (0 \le s \le p_*),$$
$$1 - \mu(p_* + s) \le \frac{1}{1 + \exp\bigl(s(\log n)^2/C_0\bigr)} \quad (0 \le s \le 1 - p_*),$$

and the endpoint cases follow by continuity.

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